

Session 6: How can I subset data and define missing values?

A Slower Introduction to R

Yea-Hung Chen, PhD, MS

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Restricting to non-
missing values

Systolic versus hypertension

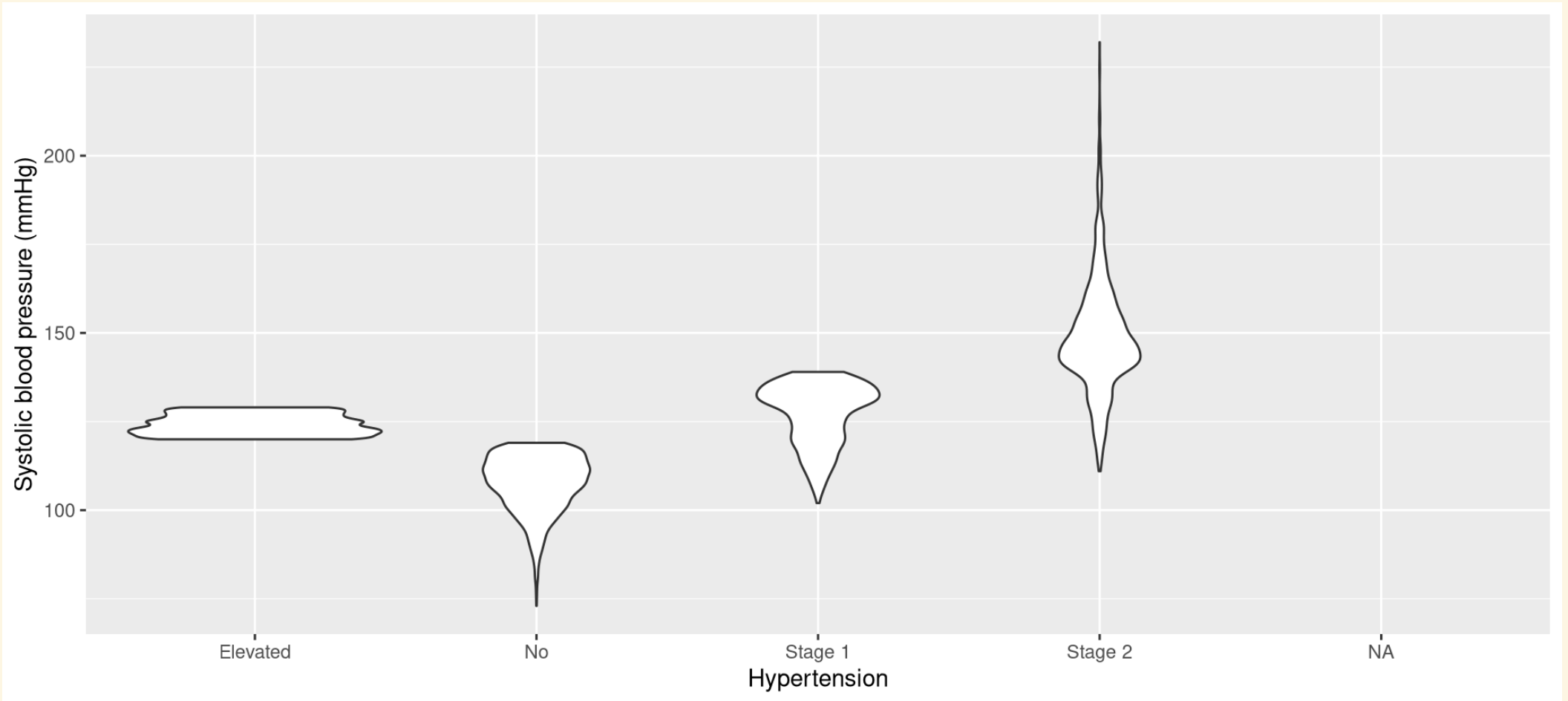
- from Exercise 1, last week:

Create a violin plot of systolic blood pressure (`bpxosy1`) versus hypertension (`hypertension`), with systolic blood pressure on the y axis. You can replace `geom_boxplot()` above with `geom_violin()`. Add labels to the axes using `labs()`. Run your code.

Systolic versus hypertension

```
ggplot(nhanes,aes(x=hypertension,y=bpxosy1))+  
  geom_violin()+  
  labs(x='Hypertension',y='Systolic blood pressure (mmHg)')
```

Systolic versus hypertension



Terminology

 Note

Synonyms of restricting include **subsetting** and filtering.

Mini data

- as a teaching example, I've created a smaller version of the NHANES data
- as always, I encourage you to **focus on watching** rather than practicing during the mini lectures
- but, here is code for `minidata` if you want to work through these examples later:

```
minidata<-nhanes[1:10,c('seqn','ridageyr','bpxosy1','bpxodi1','hypertension')]
```

Mini data

```
> minidata
      seqn ridageyr bpxosy1 bpxodi1 hypertension
1  130378      43     135     98      Stage 2
2  130379      66     121     84      Stage 1
3  130380      44     111     79           No
4  130384      43      NA     NA      <NA>
5  130385      65      NA     NA      <NA>
6  130386      34     110     72           No
7  130387      68     143     76      Stage 2
8  130388      27     130     95      Stage 2
9  130389      59     145     76      Stage 2
10 130390      31     113     78           No
```


Data for individuals not missing hypertension?

- to subset the data to individuals not missing hypertension, use `subset()` and `is.na()`:

```
> subset(minidata,!is.na(hypertension))
      seqn ridageyr bpxosy1 bpxodi1 hypertension
1  130378      43     135     98      Stage 2
2  130379      66     121     84      Stage 1
3  130380      44     111     79           No
6  130386      34     110     72           No
7  130387      68     143     76      Stage 2
8  130388      27     130     95      Stage 2
9  130389      59     145     76      Stage 2
10 130390      31     113     78           No
```

is.na()

- the values of `hypertension` are:

```
> minidata$hypertension
[1] "Stage 2" "Stage 1" "No"      NA      NA      "No"      "Stage 2"
[8] "Stage 2" "Stage 2" "No"
```

- `is.na()` shows whether the values are missing:

```
> is.na(minidata$hypertension)
[1] FALSE FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE
```

- placing an exclamation point in front reverses the values:

```
> !is.na(minidata$hypertension)
[1]  TRUE  TRUE  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE
```

subset()

```
subset(minidata,!is.na(hypertension))
```

- there are two arguments to `subset()` here:
 - `minidata`: the data I want to subset
 - `!is.na(hypertension)`: a rule defining how I want to subset
- `subset()` will keep rows that are **TRUE** in the second argument

subset()

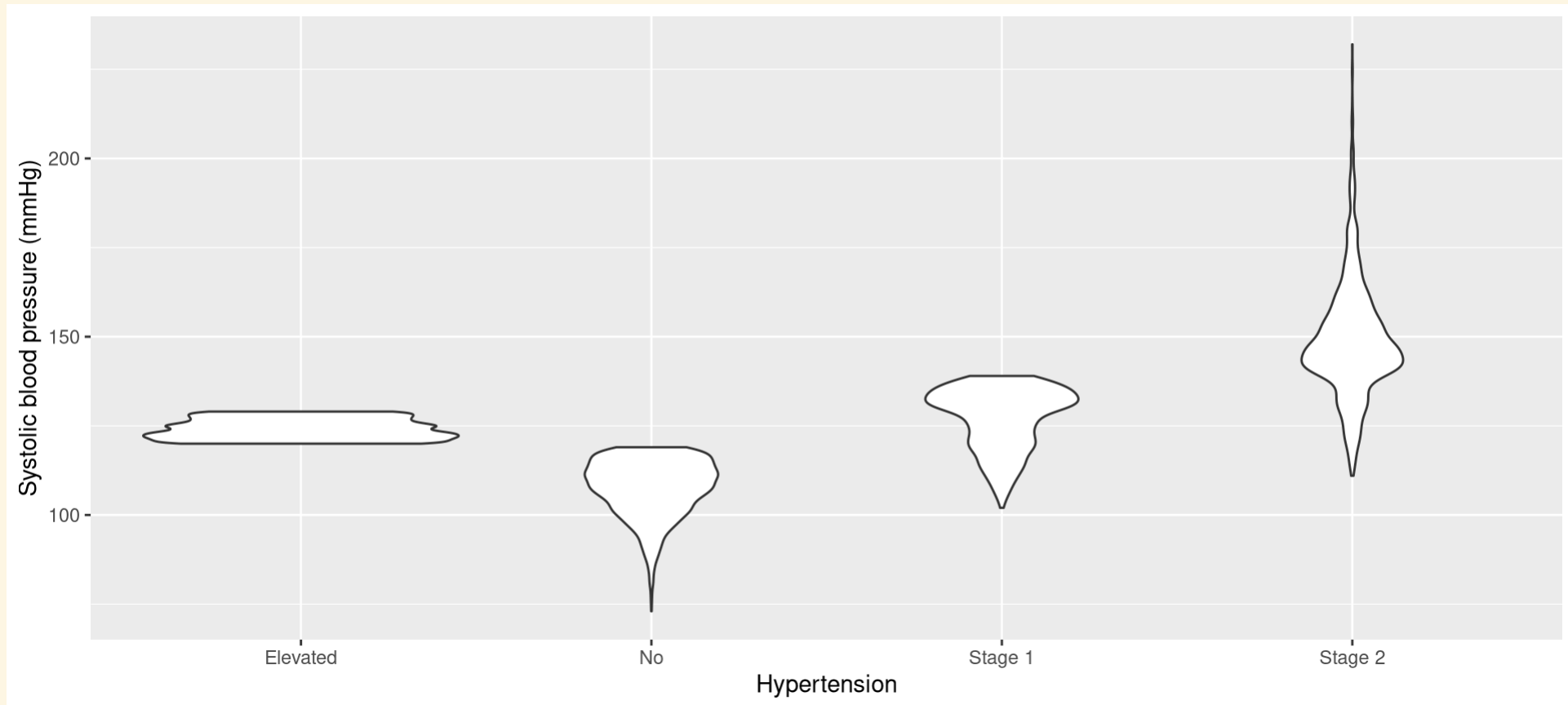
```
> subset(minidata,!is.na(hypertension))
```

- dollar-sign notation is not needed in the second argument
- in `subset()`, it is understood that any variable in the second argument is about the data in the first argument

Systolic versus hypertension

```
ggplot(subset(nhanes,!is.na(hypertension)),aes(x=hypertension,y=bpxsy1))+  
  geom_violin()+  
  labs(x='Hypertension',y='Systolic blood pressure (mmHg)')
```

Systolic versus hypertension



Notice that the groups are still out of order. We'll talk about how to handle this next

Exercise 1

Exercise 1

1. At the top of a new script, include code for loading the *ggplot2* package and importing the data. Run your code.
2. Using *ggplot2*, create density plots of systolic blood pressure (*bpxosy1*), with faceting by *asthma* (defining the rows). In *ggplot()*, define *data* to be a subset of the NHANES data that only includes individuals who are non-missing for *asthma*.

Numerical comparisons

Data for individuals 60 years of age or older?

- suppose I want to look at the data for just individuals who are 60 years of age or older

Data for individuals 60 years of age or older?

- to subset to individuals 60 years of age or older, use `subset()` and a numerical comparison:

```
> subset(minidata, ridageyr >= 60)
  seqn ridageyr bpxosy1 bpxodi1 hypertension
2 130379      66     121      84      Stage 1
5 130385      65      NA      NA          <NA>
7 130387      68     143      76      Stage 2
```

Data for individuals 60 years of age or older?

```
subset(minidata,ridageyr>=60)
```

- there are two arguments to `subset()` here:
 - `minidata`: the data I want to subset
 - `ridageyr>=60`: the rule defining how I want to subset

Numerical comparisons

```
> minidata$ridageyr  
[1] 43 66 44 43 65 34 68 27 59 31  
> minidata$ridageyr >= 60  
[1] FALSE TRUE FALSE FALSE TRUE FALSE TRUE FALSE FALSE FALSE
```

Numerical comparisons

- `==` for equal
- `!=` for not equal
- `>` for greater than
- `>=` for greater than or equal to
- `<` for less than
- `<=` for less than or equal to

Numerical comparisons

```
> minidata$ridageyr
[1] 43 66 44 43 65 34 68 27 59 31
> minidata$ridageyr==43
[1] TRUE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
> minidata$ridageyr<59
[1] TRUE FALSE TRUE TRUE FALSE TRUE FALSE TRUE FALSE TRUE
```

subset() and numerical comparisons

```
> subset(minidata, ridageyr < 59)
```

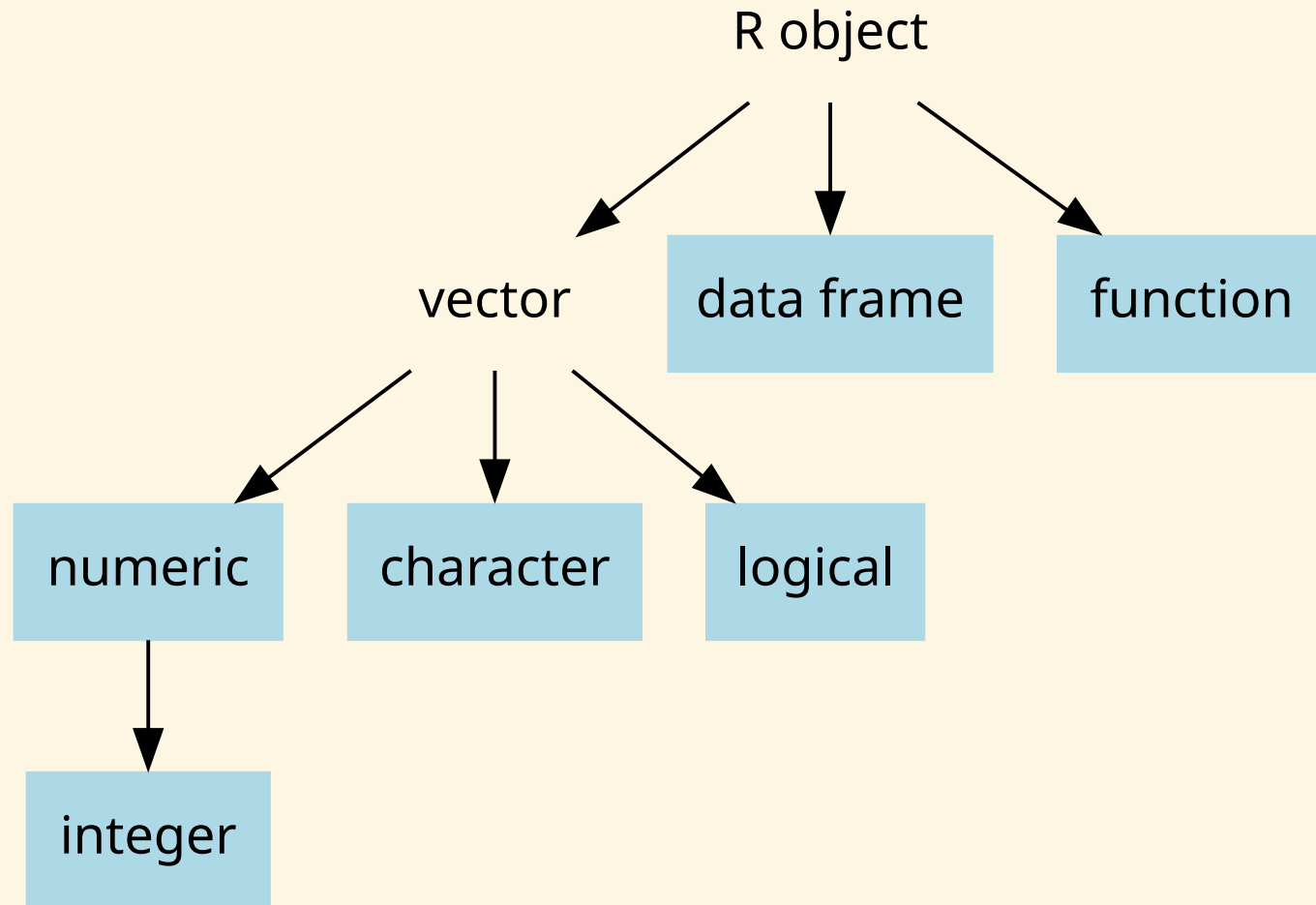
	seqn	ridageyr	bpxosy1	bpxodi1	hypertension
1	130378	43	135	98	Stage 2
3	130380	44	111	79	No
4	130384	43	NA	NA	<NA>
6	130386	34	110	72	No
8	130388	27	130	95	Stage 2
10	130390	31	113	78	No

Logical objects

Logical objects

```
> class(minidata$ridageyr==43)
[1] "logical"
> class(is.na(minidata$hypertension))
[1] "logical"
> is.vector(minidata$ridageyr==43)
[1] TRUE
> is.vector(is.na(minidata$hypertension))
[1] TRUE
```

Logical objects



Negation

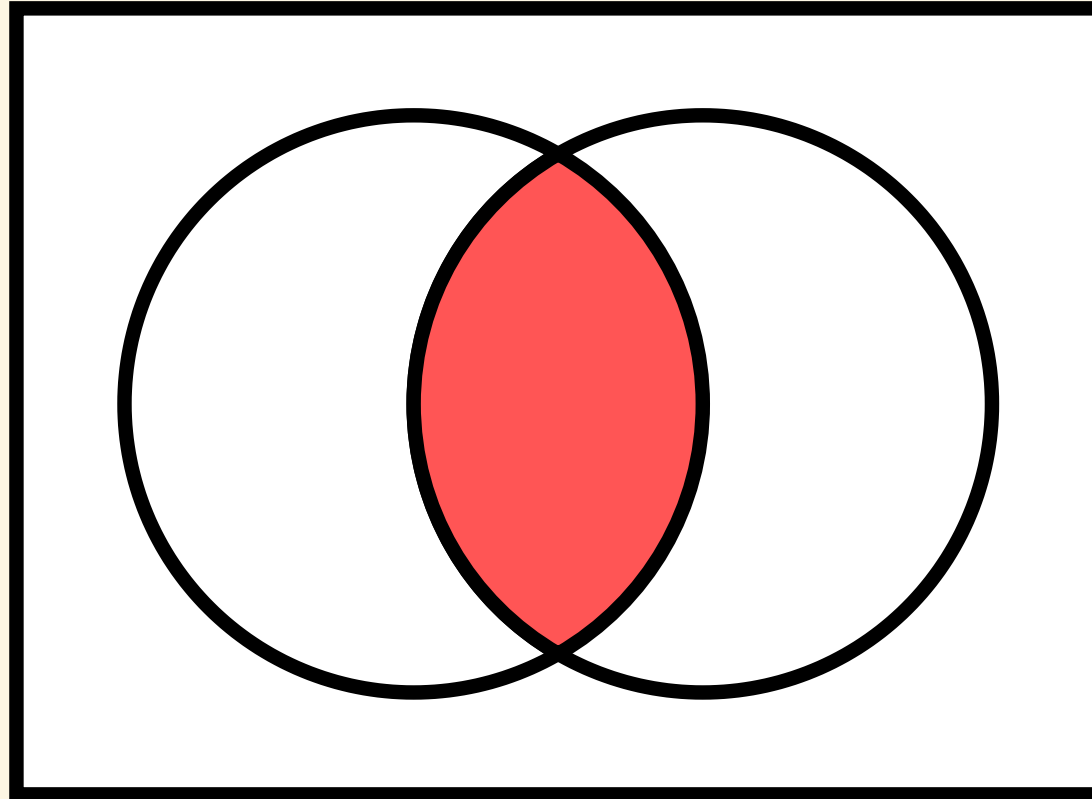
```
> minidata$ridageyr==43
[1] TRUE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
> minidata$ridageyr!=43
[1] FALSE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE
> !minidata$ridageyr==43
[1] FALSE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE
```

- to negate (reverse) a logical object, place ! at the beginning

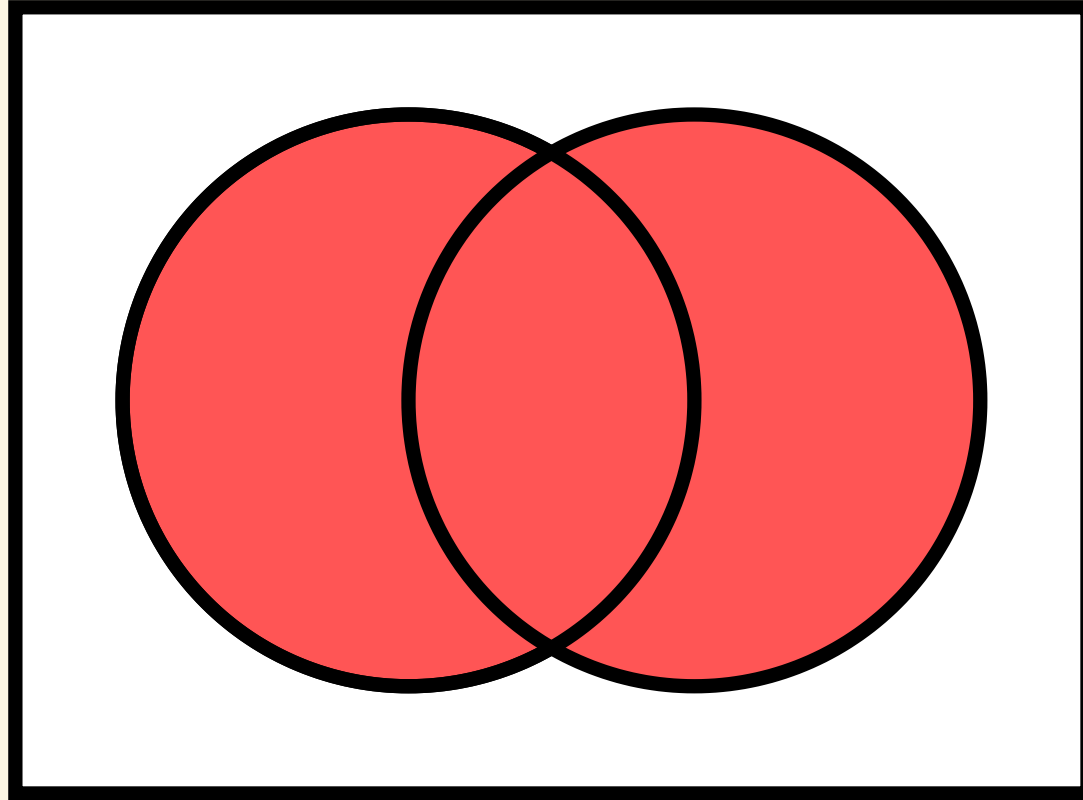
And and or

- `&` = and
- `|` = or

And



Or



And and or

- individuals without hypertension are partially defined by:

```
> minidata$bpxosy1<120 & minidata$bpxodi1<80  
[1] FALSE FALSE TRUE NA NA TRUE FALSE FALSE FALSE TRUE
```

- individuals with Stage 2 hypertension are defined by:

```
> minidata$bpxosy1>=140 | minidata$bpxodi1>=90  
[1] TRUE FALSE FALSE NA NA FALSE TRUE TRUE TRUE FALSE
```


Logical objects and math

- when combining math with logical objects, **TRUE** values are treated as **1** and **FALSE** values are treated as 0
- this can be very useful

Logical objects and math

- to count missing values:

```
> sum(is.na(minidata$hypertension))  
[1] 2
```

- or, to count the number of people within an age range:

```
> sum(minidata$ridageyr>=30)  
[1] 9
```

Exercise 2

Exercise 2

1. Copy and paste the following code for defining `ex2`. Run that code.

```
ex2<-subset(nhanes,!is.na(lbxglu)&!is.na(lbdhdd))  
ex2<-ex2[1:10,c('ridageyr','dmdborn4','dmdeduc2','lbxglu','lbdhdd')]
```

Exercise 2

2. Use `subset()` to get the subset of `ex2` individuals who are between 40 and 49 years of age (`ridageyr`), inclusive of both endpoints.
3. Use `subset()` to get the subset of `ex2` individuals who have a fasting glucose level (`lbgglu`) greater or equal to 100 mg/dL and an HDL cholesterol level (`lbdhdd`) greater or equal to 50 mg/dL.

Exercise 2

4. Count the number of `ex2` individuals who were born in the US (a `dmbdorn4` value equal to `1`) and have a high-school degree (a `dmddedu2` value equal to `3`).
5. Count the number of `ex2` individuals who are 50 years of age (`ridageyr`) or older and have a fasting glucose level (`lbxglu`) greater or equal to 100 mg/dL.

Exercise 2

6. Many of the functions for hypothesis testing support a `data` argument. This argument allows you to omit dollar-sign notation. So, these two examples are equivalent:

```
t.test(nhanes$bpxosy1~nhanes$asthma)
t.test(bpxosy1~asthma,data=nhanes)
```

Modify the `data` argument in the second example to examine the relationship between systolic blood pressure and asthma among individuals with a history of smoking (a value of `smoking` equal to `History of smoking`). Remember to omit dollar-sign notation.

Exercise 2

7. Using the `data` argument, fit a simple linear regression model for the relationship between systolic blood pressure (`bpxosy1`) and diastolic blood pressure (`bpxodi1`), with systolic blood pressure as the y variable, restricting to individuals with a history of smoking. Remember to omit dollar-sign notation.

Defining missing values

Defining new variables

- to define a new variable, use dollar-sign notation and object assignment:

```
> minidata$new<-1
```

- `new` is an arbitrary name for the new variable

Defining new variables

```
> minidata
```

	seqn	ridageyr	bpxosy1	bpxodi1	hypertension	new
1	130378	43	135	98	Stage 2	1
2	130379	66	121	84	Stage 1	1
3	130380	44	111	79	No	1
4	130384	43	NA	NA	<NA>	1
5	130385	65	NA	NA	<NA>	1
6	130386	34	110	72	No	1
7	130387	68	143	76	Stage 2	1
8	130388	27	130	95	Stage 2	1
9	130389	59	145	76	Stage 2	1
10	130390	31	113	78	No	1

Vigorous physical activity

- from NHANES, the `pad820` variable records answers to the question:

About how long {do you/does SP} do these vigorous leisure-time physical activities each time?

Vigorous physical activity

```
> table(nhanes$pad820,useNA='always')
```

1	2	3	4	5	6	7	8	10	12	15	20	25	30	32	35
12	3	6	2	36	2	1	2	108	5	190	212	30	909	1	12
40	45	46	50	55	60	70	75	90	100	105	120	150	180	240	300
70	262	1	19	3	1064	5	13	145	2	3	364	4	93	49	21
360	420	480	540	600	660	900	7777	9999	<NA>						
10	1	8	1	1	1	2	1	13	4466						

Vigorous physical activity

- from NHANES:

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 900	Range of Values	3673	3673	
7777	Refused	1	3674	
9999	Don't know	13	3687	
.	Missing	4466	8153	

Vigorous physical activity

- 7777 and 9999 are not actual minutes of vigorous physical activity
- so, we might want to recode 7777 and 9999 to be NA

Vigorous physical activity

- there are several possible ways to define missing values
- one approach is to use the `replace()` function:

```
nhanes$ltpa_vigorous<-replace(nhanes$pad820,  
                             nhanes$pad820==7777 | nhanes$pad820==9999,  
                             NA)
```


Vigorous physical activity

```
nhanes$ltpa_vigorous<-replace(nhanes$pad820,  
                             nhanes$pad820==7777|nhanes$pad820==9999,  
                             NA)
```

- there are three arguments to `replace()` here:
 - `nhanes$pad820`: the variable that we want to modify
 - `nhanes$pad820==7777|nhanes$pad820==9999`: a rule (specifically, a logical object) that indicates which values should be replaced
 - `NA`: the replacement value

Vigorous physical activity

```
nhanes$ltpa_vigorous<-replace(nhanes$pad820,  
                             nhanes$pad820==7777|nhanes$pad820==9999,  
                             NA)
```

- notice that I defined a new variable, `ltpa_vigorous`



When cleaning variables, **define new variables**, rather than replacing existing variables.

Vigorous physical activity

- to check the work, use `table()`:

```
> table(nhanes$ltpa_vigorous, useNA='always')
```

1	2	3	4	5	6	7	8	10	12	15	20	25	30	32	35
12	3	6	2	36	2	1	2	108	5	190	212	30	909	1	12
40	45	46	50	55	60	70	75	90	100	105	120	150	180	240	300
70	262	1	19	3	1064	5	13	145	2	3	364	4	93	49	21
360	420	480	540	600	660	900	<NA>								
10	1	8	1	1	1	2	4480								



Tip

Always check your work after doing any data cleaning.

Exercise 3

Exercise 3

1. `pad800` stores minutes of moderate physical activity. Values of `7777` indicate “refused” while values of `9999` indicate “don’t know” (the codebook is available [here](#)). Use object assignment and `replace()` to define a new variable for moderate physical activity, with values of `7777` and `9999` replaced with `NA`. Check your work.

Exercise 3

2. `a1q130` stores responses to the question, “During the past 12 months, on those days that you drank alcoholic beverages, on average, how many drinks did you have?” Values of `777` indicate “refused” while values of `999` indicate “don’t know.” Use object assignment and `replace()` to define a new alcohol-use variable, with values of `777` and `999` replaced with `NA`. Check your work.

Exercise 3

3. Install the *remotes* package.
4. Load the *remotes* package. You can type this code at the console.
5. Install the *basecase* package, which we will discuss next week:

```
> install_github('yea-hung/basecase')
```
6. Install the *dplyr* package, which we will discuss next week.